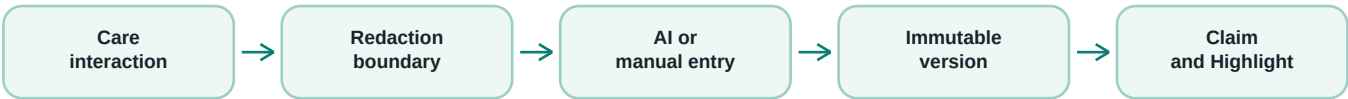


Source-linked care collaboration, not an opaque AI layer

A 72-hour synthetic-data Pilot that makes what changed, why it matters, who owns it, and where it came from visible in a single consult-ready workflow.

Architecture and trust boundary



Nightingale sits beside the EHR; it does not replace it. The EHR remains the system of record. A Highlight is a reviewable reading-order signal, not a diagnosis, risk probability, or model confidence.

Identity and tenancy Auth0 verifies browser identity. The service maps the verified subject to a provisioned user, then uses `SET LOCAL` and PostgreSQL Row-Level Security for every request.	Mutation safety The browser uses a restricted `nightingale_web` role. Security-definer procedures append versions, audit events, and outbox events in one transaction.
Evidence before interpretation Claims point to an immutable Entry Version and exact character span. View source opens that historic text and marks the precise excerpt.	Patient separation Patients receive only published summaries plus their own submitted-insight receipts. Internal Timeline entries, comments, tasks, claims, Highlights, and AI sources are excluded.

Role boundary

Patient Published instructions and own insight receipts	Staff Own Staff Notes, comments, task claim	Clinician Own Clinical Notes, Highlight review, publication	Admin Clinic-scoped governance and oversight
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A Staff member cannot create or edit a Clinician Note, and a Clinician cannot create or edit a Staff Note. These boundaries are enforced in PostgreSQL procedures and verified by the required RLS micro-tests.

One shared record, many inspectable links

The data model makes the origin and history of every surfaced signal explicit.

Entity	Relation	Linked record
Clinic	scopes	Membership, Patient, ImportanceLearning
Patient	owns	CareEntry, AIScribedSource, PatientSummary, PortalAccess
CareEntry	has	immutable EntryVersion snapshots and optional internal comments
EntryVersion	grounds	EvidenceClaim with [start,end) source span
EvidenceClaim	surfaces	Highlight with rule version and evidence state
Highlight	may trigger	Review Task and clinician review audit
AIScribedSource	cites	redacted origin linked through ai-source:

Provenance and AI

AI intake begins only after identifier redaction. The stored redacted source, generated system-authored Entry, immutable Entry Version, Claim, Highlight, rule version, and exact span form a complete inspectable chain. The workflow never silently rewrites a clinician note.

Importance learning

Importance is a bounded reading-order score. Deterministic risk rules establish the base: 95 for allergy-medication conflict or breathing difficulty, 72 for overdue renal follow-up, and 58 for an AI draft awaiting review. Clinician Accept or Pin increments a clinic-local score for that entity type by one, capped at ten. The read model shows $\min(100, \text{base importance} + \text{boost})$.

The learning mechanism cannot alter risk rules, source evidence, clinical text, patient visibility, or patient summaries. Reject and Dismiss do not inflate it. This gives a measurable effect without model-score drift.

Deliberate limits make the Pilot safer to evaluate

The system is a testable foundation for synthetic data, not a production clinical system.

First-principles decisions

- **Origin before summary:** an AI paraphrase cannot replace the source it was derived from, so Claims retain the immutable source version and span.
- **Role before convenience:** signing in never grants clinic access. Membership and patient portal access are separate capabilities.
- **History before overwrite:** edits append versions; a revert creates another version. A stale same-section write receives deterministic `VERSION\_CONFLICT`.
- **Bounded learning before personalisation:** ordering feedback remains local, capped, explainable, and non-clinical.

Assumptions and trade-offs

The Pilot assumes one clinical organisation, managed OIDC identity, and synthetic or approved de-identified input. It trades speed for inspectability: source review, explicit publication, and immutable versioning require extra steps, but make it possible to answer why a signal appeared and who changed the record. Production use would additionally require managed TLS, encryption at rest, key management, approved PHI processing, monitoring, a redaction corpus, and clinical safety review.

Verification

Required micro-tests RBAC scope, revision history, Highlight provenance, concurrent edits, and bounded importance learning all run against isolated temporary Postgres databases with the restricted web role.	Measured path The 1,000-entry synthetic Consult Glance benchmark has a recorded local P50 of 25.0 ms and P95 of 141.6 ms. It excludes Auth0 exchange and browser rendering.
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Nightingale Pilot demonstrates role-governed, source-linked, version-auditable care collaboration. All data in the demonstration is synthetic.